Aqib Shafiq

Electrical Engineer

MS Electrical Engineering, University of Azad Jammu and Kashmir

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Career Objective

Research-driven Electrical Engineer with an MS in Electrical Engineering and proven expertise in renewable energy, electronics, robotics, embedded systems, and IoT. Combining scientific research, technical innovation, and hands-on hardware development to design intelligent engineering solutions, advance emerging technologies, and contribute to high-impact research and academic excellence.

Research Interest

Smart Energy Systems, Renewable Energy Integration, Distributed Energy Resources, Smart Grids, EV Charging Optimization, Energy Management, and Power System Modelling.

Education

2020 - 2022
Masters, University of Azad Jammu and Kashmir, Muzaffarabad, Pakistan
MS in Electrical Engineering — CGPA 3.82/4.00
Thesis: design and optimisation of IoT and robotics enabled grid-tied renewables-based EV charging station

2013 - 2017
Bachelors, University of Azad Jammu and Kashmir, Muzaffarabad, Pakistan
BS in Electrical Engineering (Electronics) — CGPA 3.13/4.00
Thesis: ECG device for remote monitoring and disease prediction of cardiac patients using machine learning

Research and Work Experience

Jun 2020 - Sep 2021
Research Engineer, Muzaffarabad, Pakistan
Projects submitted for funding:
- Feasibility Study and Deployment of Solar Photovoltaic System at UAJK (submitted to HEC).
- Deployment of Solar System for University of Azad Kashmir (submitted to PSF).
- Establishment of Robotics and Automation Lab (submitted to PSF).
- Feasibility Study for Net Metering in the State of Azad Jammu and Kashmir (submitted to Ignite National Technology Fund).
- IoT-based Smart Irrigation System (submitted to Ignite National Technology Fund).

2019 - 2023
Robotics STEM Education Trainer, NEP-NIC Punjab IT Board
- Trained **4,000+ students across 30 schools** in STEM, robotics, and IoT.
- Designed and developed robotics and IoT hardware kits.
- Developed coding modules and hands-on STEM curricula.

2019 - 2023
Hardware Design Workshop Trainer
- Led practical engineering and technology workshops for students and graduates.
- Designed and developed specialized hardware training kits and functional engineering prototypes.
- Facilitated professional training in technological innovation, entrepreneurship, and startup development.

2021 - 2023
Member, Industry Advisory Board
Electrical Engineering Department, University of Azad Jammu and Kashmir

2018 - 2019

Incubatee (Team Leader), The Catalyst GIK Incubator

Ghulam Ishaq Khan Institute, Swabi, KPK, Pakistan

- Led **Cardiac Care**, a healthcare technology startup, through a one-year incubation program.
- Developed a remote ECG-based patient monitoring system integrating embedded systems and machine learning.
- Directed technical prototyping, product development, and startup innovation activities.

Aug 2017 - Sep 2017

Internee Electrical Engineer, Pakistan Engineering Services

- Gained practical exposure to electrical engineering projects, technical operations, and professional engineering practices.
- Supported engineering activities under the supervision of experienced professionals.

Jun 2017 - Jul 2017

Internee Electrical Engineer, Special Communication Organization

- Gained practical exposure to telecommunications infrastructure, electrical systems, and technical operations.
- Developed an understanding of engineering workflows, system maintenance, and organizational practices.

Research & Hardware Development Projects

Hands-on experience in engineering design, embedded systems, IoT integration, intelligent automation, and functional hardware prototype development

- **Solar PV-Based EV Charging System:** Grid-connected EV charging with renewable energy integration.
- **Wireless EV Charging Prototype:** Static wireless power transfer with IoT monitoring.
- **Smart Solar Irrigation System:** Solar-powered irrigation with sensors and embedded control.
- **Smart Substation Monitoring:** IoT-based electrical monitoring and control.
- **Healthcare Robotics:** Disinfection robot development for hospital applications.
- **Solar PV & Battery Monitoring:** IoT-enabled energy monitoring system.
- **Autonomous Robotics:** Line-following, obstacle-avoidance, and Bluetooth-controlled robots.
- **Smart Waste & Parking Systems:** IoT-enabled automation and monitoring.
- **Remote ECG Monitoring System:** Embedded hardware with machine-learning-based cardiac disease

Thesis Supervision (Jointly Supervised BS Projects)

- **Solar PV-Based Slow and Fast EV Charging Station:** Supervised the design and development of solar-powered EV charging systems supporting slow and fast charging.
- **Wireless Power Transfer for Electric Vehicles:** Supervised the design and development of a wireless EV charging system using inductive power transfer technology.

Technical Skills

Hardware & Embedded System	Circuit design, hardware prototyping, microcontrollers, robotics, IoT, and sensor interfacing.
Renewable Energy & Power Systems	Solar PV, EV charging systems, energy modeling, and techno-economic analysis.
Engineering Software & IoT Platforms	MATLAB, Simulink, HOMER Pro, HelioScope, EasyEDA, Fritzing, Arduino IDE, ThingSpeak, Blynk, and ESP8266/ESP32 Web Server.
Academic & Professional Skills	Scientific writing, research proposal development, thesis supervision, hardware prototyping mentorship, technical training, research collaboration, project leadership, and entrepreneurship.

Awards and Certificates

- **National National Innovation Award –** University of Azad Jammu & Kashmir.
- **National Business Plan Competitions –** Multiple national-level winner**.**
- **One Million AI Prompters Certification –** UAE.
- **Professional Development (CPD) –** Multiple technical workshops, Pakistan Engineering Council (PEC).
- **Technology Startup Incubation –** Team Lead, The Catalyst, GIK Institute.
- **Startup Competition Judge –** Innovation and entrepreneurship evaluation.

- **Industry Advisory Board Certification –** Electrical Engineering Department, UAJK**.**
- **Research Engineering Certification –** University-level engineering projects.
- **STEM Robotics Trainer Certification –** Training programs across 40 schools**.**
- **Robotics Summer & Winter Camps –** Certified technical trainer.
- **Engineering IoT & Robotics Training –** Graduate-level technical workshops.
- **Technical Workshop Organization –** Engineering and technology workshops**.**
- **Entrepreneurship & Career Development –** Engineering trainer and mentor.
- **Undergraduate Thesis Supervision –** Two jointly supervised BS projects.
- **Electrical Engineering Internships –** SCO and Pakistan Engineering Services (PES).

Publications

Journal Publications (8)

Impact Factor as reported at the time of publication.

Journal **Aqib Shafiq**, S. Iqbal, S. Habib, A. ur Rehman, Anis ur Rehman, et al. (2022), Solar PV-based electric vehicle charging station for security bikes: a techno-economic and environmental analysis, *Sustainability* [Impact Factor 3.889], 14(21), 13767. View online.

Journal **A. Shafiq**, et al. (2023), Transition towards a sustainable power system: MA-DA&DC framework based voltage control in high PV penetration networks, *Energy Reports* [Impact Factor 4.937], 9, 5922–5936. View online.

Journal **Aqib Shafiq**, S. Iqbal, Anis ur Rehman, et al. (2023), Integration of solar based charging station in power distribution network and charging scheduling of EVs, *Frontiers in Energy Research* [Impact Factor 3.858], 11, 1086793. View online.

Journal Anis ur Rehman, **A. Shafiq**, Z. Ullah, S. Iqbal, H. M. Hasanien (2023), Implications of smart grid and customer involvement in energy management and economics, *Energy* [Impact Factor 8.857], 276, 127626. View online.

Journal Anis ur Rehman, Z. Ullah, **A. Shafiq**, H. M. Hasanien, P. Luo, F. Badshah (2023), Load management, energy economics, and environmental protection nexus considering PV-based EV charging stations, *Energy* [Impact Factor 8.857], 281, 128332. View online.

Journal S. Iqbal, M. U. Jan, **Aqib Shafiq**, et al. (2022), Feasibility study and deployment of solar photovoltaic system to enhance energy economics of King Abdullah Campus, University of Azad Jammu and Kashmir Muzaffarabad, AJK Pakistan, *IEEE Access* [Impact Factor 3.476], 10, 5440–5455. View online.

Journal Anis ur Rehman, M. Ali, S. Iqbal, **A. Shafiq**, N. Ullah, S. A. Otaibi (2022), Artificial intelligence-based control and coordination of multiple PV inverters for reactive power/voltage control of power distribution networks, *Energies* [Impact Factor 3.252], 15(17), 6297. View online.

Journal S. Iqbal, S. Habib, M. Ali, **Aqib Shafiq**, Anis ur Rehman, et al. (2022), The impact of V2G charging/discharging strategy on the microgrid environment considering stochastic methods, *Sustainability* [Impact Factor 3.889], 14(20), 13211. View online.

Peer-Reviewed International Conference Papers (4)

Conference **Aqib Shafiq**, S. Iqbal, S. D. Ali, Anis ur Rehman, Design and cost analysis of electric vehicle charging station, 4th International Ankara Multidisciplinary Studies Congress, Ankara, Türkiye.

Conference **Aqib Shafiq**, S. Iqbal, S. D. Ali, Anis ur Rehman, Implementation of IoT based smart agriculture techniques for efficient real-time monitoring and controlling system for better yield production in Azad Kashmir, International Turkic World Congress on Science and Engineering.

Conference **Aqib Shafiq**, S. Iqbal, S. D. Ali, et al. (2022), Economic and environmental analysis for different scenarios of grid-connected solar PV-based EV charging station facility using HOMER Grid, IEEE ICETECC 2022, Jamshoro, Sindh, Pakistan, pp. 1–5. View online.

Conference Anis ur Rehman, **Aqib Shafiq**, et al. (2022), Control and coordination of multiple PV inverters in power distribution network using multi-agent deep reinforcement learning, IEEE ICETECC 2022, Jamshoro, Sindh, Pakistan, pp. 1–7. View online.